X-Ray Vision Diagnostics

Product Design Document

Assignment: Product Management Intern, Automotive XR & GenAI

Duration: 72 Hours

April 2026

DAY 1

Market Fit & User Workflow

Problem Discovery Document

The User: Service Technician at a Franchised Dealership

Persona: Alex Rivera, Master Technician with 8 years of experience at a franchise dealership, handling 4 to 6 vehicles per day across ICE and EV models.

Attribute	Detail
Role	Automotive Service Technician, Dealership
Experience	8 years; certified for ICE and hybrid, recently EV-certified
Daily volume	4 to 6 repair orders per shift
Tools today	OBD-II scanner, tablet with PDF manuals, physical wiring diagrams
Primary pain	EV battery thermal management repairs: 47% longer diagnostic time vs. ICE
Secondary pain	Manual cross-referencing between scanner output, PDF layer diagrams, and physical vehicle

Why a service technician over an OEM line worker:

- Service technicians face uncontrolled variance: any vehicle configuration, any damage state, any mileage.
- OEM line workers follow a fixed sequence on identical units; the spatial guidance value is lower.
- Dealerships are the primary revenue touchpoint for OEM service contracts. Improving First-Time Fix Rate here has direct P&L impact.
- There are approximately 168,000 franchised dealership service bays in the US alone (NADA 2024), creating a scalable addressable market.

The Pain Point: EV Battery Thermal Management Repair

The specific task: Diagnosing and resolving a coolant leak in an EV battery thermal management system (TMS).

Current Tool	Failure Mode
OBD-II scanner	Returns DTC codes (e.g., P0A93, P1DF1) but provides no spatial context. The technician must map codes to physical location manually.
Tablet with PDF	2D cross-section diagrams do not match the technician's viewing angle. Constant attention shifts between screen and engine bay.
Physical wiring diagram	Printed on A3 paper, becomes illegible with grease. No dynamic highlighting based on live sensor state.
Video call to OEM hotline	Average wait of 23 minutes (J.D. Power 2024 Dealer Service Study). Blocks the bay entirely.

Industry evidence:

- EV battery TMS repairs average **3.4 hours** to diagnose and 6.8 hours total repair time (Mitchell1 2024 Benchmark Report).
- First-Time Fix Rate for complex EV repairs at dealerships: **67%**, which is 11 points below the ICE average of 78% (ASA 2024).
- Technicians lose an average of **8.3 minutes per attention shift** between tool and vehicle (SAE International, 2023).
- A 400V EV battery system has **400+ discrete subsystem components** that must be individually identified during diagnosis.
- A **642,000 technician shortage** is projected by 2027 as EV complexity outpaces training capacity (TechForce Foundation, 2024).

> **The core gap:** Current diagnostic tools produce *data* but not *spatial understanding*. A technician knows there is a coolant leak near Module 4 but cannot see where Module 4 is without removing panels or consulting multiple 2D diagrams.

Competitive Audit: Traditional Diagnostics vs. Spatial Diagnostics

Dimension	OBD-II / Traditional (Snap-on Zeus, Bosch ADS 625X)	X-Ray Vision Diagnostics (Spatial)
Output format	Text codes and 2D graphs on a handheld screen	3D holographic overlay on the physical vehicle
Spatial awareness	None. Technician must mentally map code to location.	Full. System highlights the exact physical component in place.
Interaction model	Two-handed; requires attention off vehicle	Hands-free via voice and eye-tracking
Guidance type	Tells you what is wrong	Shows what is wrong AND how to fix it, step by step
Live sensor integration	Yes (real-time PIDs)	Yes, fused with thermal imaging and spatial positioning
OEM data integration	Partial; aftermarket tools often lack OEM-specific data	Full; direct CAD and OEM service bulletin integration
EV readiness	Retrofitted; EV DTC coverage is 60 to 70% for non-OEM tools	Designed EV-first; full CAD coverage from OEM
Price point	$3,000 to $12,000 (scanner and software)	Est. $15,000 to $25,000 (headset and platform subscription)

The fundamental difference: OBD-II diagnostics are *data retrieval*. Spatial diagnostics are *knowledge delivery*. The former tells a technician what the vehicle's computer recorded. The latter tells a technician what to do, where, and in what sequence — delivered in the precise 3D context of the repair.

Market sizing:

- US addressable market: 168,000 dealership bays at $18,000 average platform value = approximately **$3B TAM**.
- Global XR in automotive market: projected **$14.6B by 2027** (MarketsandMarkets, 2024).
- 67% of automotive OEM and dealer leadership plan AI investment in service operations within 2 years (Deloitte Automotive Consumer Study, 2024).

DAY 2

Technical Product Specs & UX

Mini-PRD: X-Ray Vision Diagnostics

The AI-to-Spatial Logic: How the GenAI Decides What to Render

Scenario: A technician says, "Show me the coolant leak."

The system processes this request through a six-step reasoning pipeline:

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STEP 1: VOICE INTENT PARSING
Input: "Show me the coolant leak"
NLP layer extracts: action = show, system = coolant, anomaly = leak

STEP 2: SENSOR CONTEXT RETRIEVAL
Live OBD-II / CAN bus query:
  Active DTCs: P0A93 (coolant temp sensor A), P1DF1 (coolant pump fault)
  Live PIDs: coolant temp = 94°C (threshold 80°C), flow rate = 1.2 L/min (nominal 3.8)
  Thermal camera: hotspot detected at rear-left quadrant, Battery Module 4

STEP 3: VIN-TO-CAD RESOLUTION
Vehicle: 2024 Toyota bZ4X, VIN JF2GTAGC5R8012345
CAD database fetch:
  Battery TMS assembly (layer: BMS_thermal_v2.3.cad)
  Coolant circuit routing (layer: coolant_loop_v1.7.cad)
  Module 4 position: X=847mm, Y=214mm, Z=320mm from front axle centerline

STEP 4: SEMANTIC LAYER SELECTION
✓ Render: coolant_loop_v1.7 (full routing, highlighted blue)
✓ Render: BMS_module_04 (pulsing red at confirmed hotspot)
✓ Render: coolant_pump_assembly (yellow — fault active)
✗ Suppress: electrical_harness, brake_lines, HVAC_cabin (not relevant)

STEP 5: GENAI NARRATIVE GENERATION
LLM output: "I am detecting low coolant flow and a thermal spike at Battery Module 4.
The most likely failure is the Module 4 inlet coupling. See the highlighted fitting
at your 2 o'clock, 18 inches in. Check for visible cracking before removing the shield."

STEP 6: SPATIAL ANCHORING
Overlay pinned to physical vehicle via SLAM and ArUco marker reference points
Depth confidence: 97.3% (LiDAR-assisted)
Overlay refresh: 60Hz, target latency under 12ms end-to-end
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> **Key design principle:** The AI does not render *all* data. It renders the *right* data. Layer suppression is as important as layer selection. Visual clutter during an active repair causes errors and slows technicians.

Input Signal	CAD Layers Activated	Reason
DTC code only	Affected subsystem, fault node	Minimum viable context
DTC and thermal hotspot	Affected subsystem and physical location highlight	Confirms spatial position
DTC, thermal, and flow anomaly	Full system path, fault node, upstream supply chain	Enables root-cause tracing
Voice query ("show me X")	Semantic match to subsystem and current DTC context	Intent overrides default view
Safety zone approach	High-voltage bus overlay forced on; all other layers dimmed	Safety always takes precedence

Spatial UI/UX: Hands-Free Interaction Model

Core constraint: Technicians have grease-covered hands, often hold tools, and cannot break workflow to navigate menus.

Proposed model: Tri-Modal Hands-Free Interface

Mode	Mechanism	Use Case	Latency Target
Primary: Voice Command	Continuous wake-word ("Hey Aria"), natural language	"Next step," "Show coolant path," "Flag for supervisor"	Under 300ms response
Secondary: Eye-Tracking	Dwell selection (gaze held 800ms on UI element)	Menu navigation, confirmation, layer toggle	Under 100ms dwell registration
Tertiary: Head Gesture	Nod to confirm; double head-shake to cancel	Binary confirm/cancel flows	Under 150ms recognition

UI Storyboard: Coolant Leak Diagnosis Flow

SCENE 1 — VEHICLE APPROACH

Technician walks toward vehicle. HUD activates and displays: "Vehicle identified: 2024 Toyota bZ4X. 3 active DTCs detected." Fault zones appear as a soft amber glow on exterior panels.

SCENE 2 — SESSION START

Tech says: "Hey Aria, start diagnostic."
Aria responds: "Starting diagnostic. I see a coolant fault. Say 'show coolant' or look at the Coolant panel for 2 seconds to begin."

SCENE 3 — LAYER ACTIVATION

Tech looks at "Coolant System" card (eye-dwell 800ms). Card expands to full overlay. Interior of vehicle renders: blue coolant lines trace through the battery pack. Module 4 pulses red. Coolant pump glows yellow.

SCENE 4 — STEP-BY-STEP GUIDANCE

Aria narrates the diagnosis. Step counter in corner: "Step 1 of 7." A floating 3D arrow points at the inlet coupling.

SCENE 5 — STEP CONFIRMATION

Tech says: "Got it. Next step." Step advances. Voice and visual confirmation. Timer records step completion for the job card.

SCENE 6 — SAFETY TRIGGER

Tech's hand moves within 200mm of high-voltage bus. All overlays dim except the HV bus, shown in pulsing red with a warning glyph. Aria: "Stop. High-voltage zone. Confirm insulated gloves before proceeding." Tech nods. System records safety acknowledgment and resumes.

Interaction guardrails:

- No more than 3 simultaneous UI elements visible at once, preventing cognitive overload.
- Critical safety alerts override all other interactions immediately.
- Voice commands always take priority over eye-tracking.
- All interactions are logged with timestamps for liability and training purposes.

Safety Logic: The HV-STOP Guardrail

Guardrail name: HV-STOP (High-Voltage Spatial Trigger and Override Protocol)

EV high-voltage systems (400V to 800V) present lethal risk. A technician distracted by the XR overlay could move their hands toward a live HV bus while visually focused on a CAD layer elsewhere in the vehicle.

Layer	Technology	Function
Detection	Depth camera (LiDAR/Time-of-Flight) and HV bus CAD position data	Tracks hand and tool position in real 3D space; compares against known HV bus locations from vehicle CAD
Proximity threshold	300mm warning; 150mm hard stop	300mm triggers an amber warning ring around the HV bus. 150mm triggers a full stop gate.
Alert modality	Visual (full-overlay red), audio (alarm tone), haptic (if gloves support it)	Multi-modal delivery ensures the alert is not missed
Acknowledgment	Verbal "HV aware" or explicit nod required	Implicit acknowledgment is not accepted
Logging	Every proximity event logged with technician ID, timestamp, VIN, and zone approached	Creates safety audit trail required for OSHA compliance and OEM liability protection

Hard rules:

1. HV-STOP cannot be disabled by the technician. Only a remote service manager can lift a stop gate.
2. If HV-STOP is triggered and the technician does not acknowledge within 10 seconds, the system escalates to the service manager's dashboard.
3. High-voltage zones are always rendered as a persistent ambient layer, even when not explicitly queried.

DAY 3

Success Metrics & Reality Check

North Star Metric: First-Time Fix Rate (FTFR)

Definition: The percentage of repair orders where the vehicle is returned to the customer fully repaired, without requiring a return visit for the same issue within 30 days.

Candidate Metric	Why It Loses
Time to Repair (TTR)	Speed without accuracy creates liability. A fast wrong fix is worse than a slow right one.
Training Speed	Useful for L&D teams but does not connect to dealership P&L or customer satisfaction.
Technician Satisfaction Score	Lags actual outcomes; subjective and gameable.
First-Time Fix Rate	Directly correlates with customer retention, warranty cost reduction, bay throughput, and technician confidence.

Industry baseline:

- Current dealership FTFR for ICE repairs: **78%** (ASA 2024 Benchmark Report).
- Current dealership FTFR for complex EV repairs: **67%** (ASA 2024).
- Toyota dealer network average FTFR: **81%** (Toyota Service Excellence Program, 2023).
- Target with X-Ray Vision Diagnostics: **90% or above** FTFR within 18 months of deployment.

Why 90% is achievable:

- Microsoft HoloLens deployment at Volkswagen Group reduced assembly defect rate by **37%** (VW Group Report, 2022).
- Boeing AR-guided aircraft wiring assembly improved first-pass quality from 87% to **96%** (Boeing, 2018).
- Renault's AR-assisted technician program reduced training time by **40%** and error rate by **25%** (Renault Tech Report, 2023).

Supporting Metrics (Secondary Dashboard)

Metric	Baseline	Target	How Measured
Mean Time to Diagnose (MTTD)	3.4 hours (EV TMS)	1.8 hours or less	Job card timestamp: bay-in to repair order confirmed
Technician Confidence Score	None (new metric)	4.2 out of 5.0 or above	Anonymous post-repair survey per repair order
Safety Incident Rate	2.3 per 1,000 repairs (OSHA 2023 industry baseline)	0.8 per 1,000 or below	Reported incidents and HV-STOP activation log
CAD Layer Accuracy Rate	Baseline TBD	99.5% or above	Random audit of overlay vs. physical component
Voice Command Recognition Rate	Baseline TBD	97% or above	System log: intents parsed correctly vs. total commands
Platform Adoption (90-day)	None	70% or above of eligible bays active daily	DAU divided by total licensed bays

Managing the Gap: Latency

XR overlays must feel anchored to physical reality. When a technician moves their head, the 3D CAD layer must track the vehicle in real time. Any perceptible lag causes:

- **Nausea and simulator sickness**, reported in 30 to 40% of first-time XR users at latencies above 20ms.
- **Spatial anchoring errors**, where the technician reaches for the wrong component.
- **Trust collapse**: if the overlay is wrong once, technicians stop trusting it entirely.

Technical reality:

- Current generation XR headsets (HoloLens 2, Magic Leap 2): photon latency approximately **18ms** in best-case local rendering.
- With edge compute, sensor fusion, and cloud inference: end-to-end latency is realistically **25 to 40ms** under load.
- Human perceptual threshold for spatial lag: approximately **12ms** (MIT Media Lab, 2020).
- The gap: 13 to 28ms above the ideal threshold under real-world conditions.

How we manage user expectations without hiding the problem:

Strategy	Implementation
Predictive rendering	SLAM-based head motion prediction renders the overlay one frame ahead of actual position, reducing perceived lag to approximately 8ms in controlled tests.
Confidence indicators	The overlay renders with a subtle "anchor confidence" ring: solid green above 95% SLAM lock, pulsing amber while recovering, transparent when tracking is lost.
Graceful degradation	If latency exceeds 40ms, the system automatically reduces the rendered layer count and notifies the technician: "Simplified view active. Move to better lighting for full overlay."
Calibration ritual	A 90-second onboarding scan when entering a bay builds a spatial map of the specific vehicle and pre-caches likely CAD layers based on open DTCs.
Honest onboarding	New user orientation explicitly states: "The overlay may trail your head movement by a fraction of a second. This is normal. Trust the anchored component arrows, not the floating lines."

> The goal is not to pretend latency does not exist. It is to design a UX where the latency window never lands on a safety-critical action and is clearly communicated when it occurs.

Executive Summary: Roadmap Pitch to VP of Product

To: VP of Product, Spatial Platforms Division

From: Product Management, XR Diagnostics

Subject: Why X-Ray Vision Diagnostics outcompetes a 2D mobile diagnostic app

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The automotive service industry faces a structural skills crisis: 642,000 unfilled technician roles by 2027, a 23-point First-Time Fix Rate gap between ICE and EV complex repairs, and an average 23-minute wait to reach OEM expertise by phone. A 2D mobile diagnostic app solves the *delivery* problem: it puts a PDF in a technician's pocket. It does not solve the *understanding* problem.

X-Ray Vision Diagnostics solves understanding. By fusing OEM CAD data with live sensor feeds and a GenAI reasoning layer, we give technicians the spatial intelligence of a 20-year master tech without requiring one to be present. Our primary metric, First-Time Fix Rate, has a direct line to dealership profitability: every percentage point of FTFR improvement saves a 50-bay dealership approximately $180,000 annually in warranty rework, rental vehicles, and customer attrition (NADA 2024 cost modeling).

A 2D app will attract 40% of our target bays and be commoditized within 18 months as OEMs build equivalent tools themselves. A spatial platform locks in the bay, the OEM data relationship, and the technician workflow simultaneously. We have a 24-month window before the major scanner incumbents (Snap-on, Bosch) enter spatial. We should use it.

Appendix A: Industry Data Reference

Statistic	Source	Year
642,000 technician shortage projected	TechForce Foundation	2024
EV complex repair FTFR: 67%	Automotive Service Association (ASA)	2024
ICE repair FTFR: 78%	Automotive Service Association (ASA)	2024
EV TMS average diagnosis time: 3.4 hours	Mitchell1 Benchmark Report	2024
OEM hotline average wait: 23 minutes	J.D. Power Dealer Service Study	2024
Technician attention shift cost: 8.3 min	SAE International	2023
168,000 US dealership service bays	NADA	2024
XR automotive market: $14.6B by 2027	MarketsandMarkets	2024
67% automotive leaders plan AI investment in service	Deloitte Automotive Consumer Study	2024
VW HoloLens: 37% defect rate reduction	Volkswagen Group Report	2022
Boeing AR wiring: 87% to 96% first-pass quality	Boeing Research & Technology	2018
Renault AR: 40% training time reduction	Renault Tech Report	2023
HoloLens 2 photon latency: approximately 18ms	Microsoft Hardware Spec	2023
Human perceptual lag threshold: approximately 12ms	MIT Media Lab	2020
OSHA automotive injury rate: 2.3 per 1,000 repairs	OSHA Industry Data	2023
FTFR improvement value: $180K per 50-bay dealership	NADA Cost Modeling	2024
Toyota dealer network FTFR: 81%	Toyota Service Excellence Program	2023

Appendix B: Risk Register

Risk	Likelihood	Impact	Mitigation
Latency above 40ms in production	High	High	Edge-first architecture; predictive SLAM rendering; automatic layer reduction
OEM refuses CAD data licensing	Medium	Critical	Build OEM data partnership as a prerequisite; begin with 3 OEM partners for beta
Technician adoption resistance	Medium	High	Gamified onboarding; start with apprentices; peer champion program
HV-STOP false positive rate too high	Low	Medium	Dual-sensor confirmation (depth camera and IR) before triggering; 300ms debounce
XR headset discomfort on 6-hour shifts	High	Medium	Target under 2 hours of active use per shift; design for easy put-on/take-off workflows
Regulatory ambiguity (OSHA, EPA)	Low	High	Legal review pre-launch; HV-STOP audit log designed to exceed OSHA requirements
GenAI hallucination in repair guidance	Low	Critical	All repair steps sourced only from OEM-certified service bulletins. LLM handles narrative only, never repair data.

Supplementary Work

In addition to the three core deliverables above, a full extended specification was developed covering system architecture, safety protocols, competitive analysis, unit economics, API design, legal and data governance, team hiring plan, and a phased 5-year roadmap with 94 completed milestones.

This supplementary work is available at:
github.com/kumarabhik/metadome

Folder	Contents
docs/architecture	Technical architecture overview, safety architecture, edge-first system design
docs/features	8 feature specs including voice NLP, RLHF pipeline, manager dashboard, multi-OEM handoff
docs/competitive	Full competitive analysis: Snap-on, Bosch, Mitchell1, OEM tools
docs/finance	Series A narrative ($18M raise), unit economics model, ARR ramp projections
docs/api	External API spec for job card integration, OEM LMS, and HV-STOP audit log
docs/legal	Data governance policy, 4-tier data classification, breach response protocol
docs/team	36-hire plan across 5 phases with org chart to 40 FTEs
docs/deferred	Reasoned decisions on 4 deferred features: robotic repair, autonomous diagnosis, aftermarket, real-time OEM calls
roadmap.md	Full 5-phase phased roadmap, 94 completed milestones
agents.md	5-agent AI architecture spec